THEOREM 5.1. *Let $a \geq 1$, $m, e, f \geq 2$ be integers such that $a < m$, $(e-1) \cdot (f-1) < m < ef$. Then the problem*

$$m \mid ax + y, \qquad 0 < x < e, \ |y| < f, \ y \neq 0 \tag{5.1}$$

is solvable in integers x, y if and only if $d = \mathrm{GCD}(a, m) < f$. Furthermore, assuming that this is the case, let

$$\frac{p_0}{q_0} = \frac{0}{1}, \frac{p_1}{q_1}, \ldots, \frac{p_N}{q_N} = \frac{a/d}{m/d}, \qquad q_N \geq \frac{m}{f-1} > e-1,$$

be the continued fraction approximations of a/m, and choose n such that $q_n < e \leq q_{n+1}$. Then $x_1 = q_n$, $y_1 = mp_n - aq_n$ is a solution for (5.1). *The set of all solutions for* (5.1) *exclusively either consists of $\lambda x_1, \lambda y_1$, $1 \leq \lambda < \min(e/x_1, f/|y_1|)$, or else consists of x_1, y_1 and x_2, y_2 with $y_1 y_2 < 0$. In the latter case, we can determine x_2, y_2 from p_{n-1}/q_{n-1} or p_{n+1}/q_{n+1} in $O(\log^2 m)$ steps.*

Before we can prove this theorem, we need to establish a lemma from the theory of continued fraction approximations. Following Hardy and Wright [8, Section 10], we denote a continued fraction by

$$[a_0, a_1, \ldots, a_n, \ldots] = a_0 + \frac{1}{[a_1, \ldots, a_n, \ldots]}.$$

The nth convergent is given by

$$\frac{p_n}{q_n} = [a_0, a_1, \ldots, a_n]$$

and satisfies $p_n q_{n-1} - p_{n-1} q_n = (-1)^{n-1}$. Notice that

$$[a_0, a_1, \ldots, a_n, 1] = [a_0, a_1, \ldots, a_n + 1], \tag{5.2}$$

but this is the only amibiguity possible for the *simple* continued fraction expansion of a real number where a_i, $i > 0$, are positive integers.

LEMMA 5.1 ([8, Theorem 172]). *If*

$$x = \frac{PZ + R}{QZ + S},$$

where $Z \geq 1$ and P, Q, R, and S are integers such that

$$Q > S > 0, \qquad PS - QR = \pm 1,$$

then R/S and P/Q are two consecutive convergents in the simple continued fraction expansion of x, provided we choose the left-hand side of (5.2) *to resolve ambiguity.*

Proof. Consider the continued fraction expansion

$$\frac{P}{Q} = [a_0, a_1, \ldots, a_n] = \frac{p_n}{q_n}.$$

From (5.2) it follows that we may choose n even or odd as we need it. Now let n be such that

$$PS - QR = (-1)^{n-1} = p_n q_{n-1} - p_{n-1} q_n.$$

Now $\mathrm{GCD}(P, Q) = 1$ and $Q > 0$, hence $P = p_n$ and $Q = q_n$, and therefore

$$p_n(S - q_{n-1}) = q_n(R - p_{n-1}).$$

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This implies that $q_n \mid (S - q_{n-1})$, which by virtue of $q_n = Q > S > 0$ and $q_n > q_{n-1} > 0$ is only possible if $S - q_{n-1} = 0$. Therefore, $R = p_{n-1}$ and $S = q_{n-1}$, and

$$x = \frac{p_n Z + p_{n-1}}{q_n Z + q_{n-1}} \quad \text{implies} \quad x = [a_0, a_1, \ldots, a_n, Z]. \quad \square$$

Proof of Theorem 5.1. Since $m \mid ax + y$, there exists an integer z such that $y = mz - ax$ and thus $d \mid y$, which implies $d \le |y| < f$. For p_n/q_n we have, as in the proof of Lemma 3.1,

$$\left|\frac{a}{m} - \frac{p_n}{q_n}\right| \le \frac{1}{q_n q_{n+1}} \le \frac{1}{q_n e};$$

therefore, $|y_1| = |aq_n - mp_n| \le m/e < f$. Notice that $y_1 \ne 0$ because $a/m \ne p_n/q_n$ for $q_n < q_N = m/d$.

We prove next that if $\hat{x}_1, \hat{y}_1$ also solve (5.1), and $y_1\hat{y}_1 > 0$, then $y_1/x_1 = \hat{y}_1/\hat{x}_1$. Since $\hat{x}_1(ax_1 + y_1) \equiv x_1(a\hat{x}_1 + \hat{y}_1) \equiv 0 \bmod m$, we have $y_1\hat{x}_1 - \hat{y}_1 x_1 \equiv 0 \bmod m$. But $|y_1\hat{x}_1|, |\hat{y}_1 x_1| \le (e-1)(f-1) < m$, hence $y_1\hat{x}_1 = \hat{y}_1 x_1$, which is our claim. Nonetheless, it can happen that $\mathrm{GCD}(x_1, y_1) = g \ne 1$, and therefore the assertion $\hat{x}_1 = \lambda x_1$, $\hat{y}_1 = \lambda y_1$ needs proof. First we note that

$$ax_1 + y_1 = mp_n \quad \text{and} \quad \mathrm{GCD}(p_n, x_1) = \mathrm{GCD}(p_n, q_n) = 1.$$

Thus,

$$a\lambda\frac{x_1}{g} + \lambda\frac{y_1}{g} = \frac{\lambda m p_n}{g} \equiv 0 \bmod m$$

if and only if $g \mid \lambda$, since $\mathrm{GCD}(g, p_n) = 1$.

Assume now that there exists a second solution x_2, y_2 to (5.1) such that $y_1 y_2 < 0$. Let $ax_1 + y_1 = mz_1$, $ax_2 + y_2 = mz_2$. Again, by multiplying the first equation with x_2, the second with x_1 and subtracting, we get $x_1 y_2 - x_2 y_1 \equiv 0 \bmod m$. Since $|x_1 y_2|, |x_2 y_1| < m$ and $y_1 y_2 < 0$, we must have $|x_1 y_2 - x_2 y_1| = m$. Thus, $m|x_1 z_2 - x_2 z_1| = |x_1 y_2 - x_2 y_1| = m$, which implies that

$$|x_1 z_2 - x_2 z_1| = 1. \tag{5.3}$$

One immediate conclusion from (5.3) is that no solutions proportional to either x_1, y_1 or x_2, y_2, the only other possible solutions (as shown before), can occur. For otherwise, for any of these solutions, say $\hat{x}$, $\hat{y}$, one has $\mathrm{GCD}(\hat{x}, \hat{z}) \ne 1$, where $\hat{z}$ is the corresponding, also proportional, multiplier of m.

It is harder to show how this second solution x_2, y_2 can be computed in case it exists. Surprisingly, this alternate solution can arise in two different ways. Without loss of generality, let us assume that $x_2 < x_1$. Notice that by this assumption we now only know that either x_1, y_1 or x_2, y_2 is the solution found as stated in the theorem.

Case 1: $|y_2| > |y_1|$. Let $Z = |y_2/y_1| > 1$. Then $Zmz_1 = Zax_1 + Zy_1 = Zax_1 - y_2$, hence $Zmz_1 + mz_2 = Zax_1 + ax_2$, or

$$\frac{a}{m} = \frac{Zz_1 + z_2}{Zx_1 + x_2}.$$

All conditions to Lemma 5.1 with $P = z_1$, $Q = x_1$, $R = z_2$, and $S = x_2$ are now satisfied (refer in particular to (5.3)), and we can conclude that z_2/x_2 and

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z_1/x_1 must be consecutive convergents to a/m. Therefore, $x_2 = q_{n-1}$ and $y_2 = mp_{n-1} - aq_{n-1}$.

Case 2: $|y_2| \le |y_1|$. Consider for an integer $k \ge 0$,

$$x_3 = x_1 + kx_2, \quad y_3 = y_1 + ky_2, \quad z_3 = z_1 + kz_2,$$

such that

$$|y_2| \ge |y_3| \quad \text{and} \quad \operatorname{sign}(y_3) = -\operatorname{sign}(y_2).$$

Now

$$0 < x_2 < x_3, \quad \frac{a}{m} = \frac{Zz_3 + z_2}{Zx_3 + x_2}, \quad Z = \left|\frac{y_2}{y_3}\right| \ge 1, \qquad z_3x_2 - z_2x_3 = \pm 1,$$

and Lemma 5.1 applies again. Therefore, z_2/x_2 and z_3/x_3 are consecutive continued fraction approximations. Notice that $Z = 1$ is possible if and only if z_2/x_2 is the second to the last convergent of a/m. In that case,

$$\frac{z_3}{x_3} = [0, a_1, \ldots, a_N - 1] \quad \text{where } \frac{a}{m} = [0, a_1, \ldots, a_N].$$

In order to compute x_1 and y_1, we must find k. Since $|y_1| = |y_3 - ky_2|$ is monotonically increasing with k, we choose the smallest k such that $x_3 - kx_2 \le e - 1$. In other words, the only possible value is

$$k = \left\lceil \frac{x_3 - e + 1}{x_2} \right\rceil.$$

Observe that $x_1 = x_3 - kx_2 \ne x_2$, since $\mathrm{GCD}(x_2, x_3) = 1$, and

$$ax_1 + y_1 \equiv 0 \bmod m \quad \text{for } y_1 = y_3 - kx_2. \quad \square$$

The following example shows that all three types of solutions, namely either a single one, or two, or a family of proportional solutions do occur.

Examples. Let $m = 11$, $e = f = 4$, $a = 7$. The continued fraction expansion of $7/11$ is $[0, 1, 1, 1, 3]$ and the convergents are $0/1$, $1/1$, $1/2$, $2/3$, and $7/11$. Hence, $x_1 = 3$, $y_1 = 2 \cdot 11 - 7 \cdot 3 = 1$, and $x_2 = 2$, $y_2 = 1 \cdot 11 - 7 \cdot 2 = -3$ are the only solutions for (5.1) in this case. For $a = 2$, the only solution is $1 \cdot 2 - 2 \equiv 0 \bmod 11$, but for $a = 1$, there are three solutions, $1 \cdot 1 - 1 \equiv 2 \cdot 1 - 2 \equiv 3 \cdot 1 - 3 \equiv 0 \bmod 11$.

Next, let $m = 244$, $a = 47$, $e = 7$, and $f = 39$. The first three convergents are $0/1$, $1/5$, and $5/26$. Thus, $x_2 = 5$ and $y_2 = 244 \cdot 1 - 47 \cdot 5 = 9$. The second solution is obtained from $k = \lceil (26 - 7 + 1)/5 \rceil = 4$ as $x_1 = 26 - 4 \cdot 5 = 6$ and $y_1 = 38$.

Finally, let $m = 56$, $a = 21$, $e = 7$, and $f = 9$. The continued fraction expansion of $21/56$ is $[0, 2, 1, 2]$ and the convergents are $0/1$, $1/2$, $1/3$, and $3/8$. Thus, $x_2 = 3$, $y_2 = -7$. We obtain $z_3/x_3 = [0, 2, 1, 1] = 2/5$, $k = \lceil (5 - 7 + 1)/3 \rceil = 0$, $x_1 = 5$, and $y_1 = 7$. $\square$

One application of Theorem 5.1, described by Wang [31] and others, is to recover rational numbers from their modular representations. Set $y/x \in \mathbf{Q}$ and suppose we have found bounds e and f for the denominator and numerator, respectively. Then, having computed $a = yx^{-1} \bmod m$, $\mathrm{GCD}(x, m) = 1$, $(e-1)(f-1) < m$, we can find y/x by continued fraction approximation. Unfortunately, we may get two possible fractions y_1/x_1, y_2/x_2 of opposite sign. One way to resolve the ambiguity is to choose the e twice the bound of the denominator and select the solution with

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$x < e/2$, $|y| < f$. Wang, in fact, chooses m such that $\lceil\sqrt{m/2}\rceil$ is a bound for both numerator and denominator. In this application, the existence of a solution is assumed and Thue's theorem does not come into play.

We now apply Theorem 5.1 to the problem of factoring a rational prime p in the UFD O_d, d fixed. We again must precondition our algorithm by factoring all rational primes smaller than $2\sqrt{|D|}$ ($\sqrt{D}$ suffices for $D > 0$). First consider $p \nmid d$. From Section 2 we know that $p < \sqrt{D}$ factors if and only if $(D/p) = +1$. In that case, a prime factor of p is $\pi = \mathrm{GCD}(p, l + \sqrt{D})$, where $l^2 \equiv D$ modulo p. We can compute l by either the Tonelli-Shanks algorithm (cf. D. Knuth [14, Section 4.6.2, Exercise 15]) or by Schoof's algorithm [26]. The latter is deterministic and runs in $O(\log^8 p)$ steps, since $|d|$ is fixed. The GCD algorithm of Section 3 can give us the wanted factor π, but in this special case, Theorem 5.1 can be applied to our advantage. Let

$$e = \left\lceil \sqrt{p/\sqrt{|D|}} \right\rceil \quad \text{and} \quad f = \left\lceil \sqrt{p\sqrt{|D|}} \right\rceil .$$

Then, $(e-1)(f-1) < p < ef$ and $2 \le e \le f$ for $p > \sqrt{|D|}$. Then we compute the continued fraction approximation to l/p and get integers x, y such that $p \mid yl + x$, $0 < y < e$, $|x| < f$. This solution satisfies

$$0 \equiv (x + ly)(x - ly) \equiv x^2 - Dy^2 \bmod p,$$

thus there exists an integer q with $x^2 - Dy^2 = qp$. By our bounds we get

$$x^2 - Dy^2 < p\sqrt{|D|} - D\frac{p}{\sqrt{|D|}} = 2\sqrt{|D|}p \quad \text{for } D < 0,$$
$$|x^2 - Dy^2| < \max(x^2, Dy^2) < \sqrt{D}p \quad \text{for } D > 0,$$

thus $|q| < 2\sqrt{|D|}$.

In the other case $p \mid d$, $p > \sqrt{|D|}$, we have $x^2 - Dy^2 = qp$ with $x = 0$, $y = 1$ and $q < \sqrt{D}$. In both cases the factorization of q into primes $q = \gamma_1 \cdots \gamma_k$ in O_d is already known. Since O_d is a UFD and

$$(x + \sqrt{D}y)(x - \sqrt{D}y) = \gamma_1 \cdots \gamma_k p,$$

γ_i must divide $x + \sqrt{D}y$ or $x - \sqrt{D}y$. Let γ be a maximum product of γ_i such that γ divides $x + \sqrt{D}y$. Then q/γ divides $x - \sqrt{D}y$, and the prime factors of p are then $(x + \sqrt{D}y)/\gamma$ and $(x - \sqrt{D}y)/(q/\gamma)$. To prove this, we only have to show that neither quotient is a unit. This follows from the fact that the division can only decrease the norm of the dividend. If one quotient became a unit, the other one would have to have the norm of their product, p^2, which is larger than its original norm qp.

We now discuss how to factor $\xi \in O_d$ with O_d a unique factorization domain. We first factor $\mathbf{N}\xi = p_1 \cdots p_k$ over the integers. If $(d/p_i) = +1$ or p_i divides d, we split $p_i = \pi_i\overline{\pi}_i$ by the algorithm discussed above. We thus obtain a factorization $\xi\overline{\xi} = \pi_1 \cdots \pi_l$, and it remains to trial-divide ξ by π_i, $1 \le i \le l$, to determine which are its prime factors.

6. Conclusion. We have described algorithms for taking the greatest common divisor and computing the prime factorization of numbers in quadratic fields with

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unique factorization. The methods also apply to computing canonical representations of unions of ideals in quadratic number rings without unique factorization. Our algorithms are of polynomial running time provided we fix the discriminant. We have also shown how to reduce factorization in quadratic number rings with unique factorization to rational integer factorization. If the discriminant is large, say of order 10^{10}, our algorithms unfortunately become impractical. Future investigations will focus on how to treat these cases efficiently.

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